Fundamentals 26

Siddharth K

May 2025

Problem 26.1

Let $x \in X$; we wish to find countable $\mathcal{U}$ of neighborhoods of x where each neighborhood contains some $U \in \mathcal{U}$. Every open set is a union of elements of a countable basis $\mathcal{B}$, so every neighborhood of x contains an element of $\mathcal{B}$. So we can let $\mathcal{U} = \{B \in \mathcal{B} : x \in B\}$.

Problem 26.2

We wish to prove that $\bar{\mathbb{Q}} = \mathbb{R}$. Let $\mathbb{Q} \subseteq C$ where $C \subseteq \mathbb{R}$ is closed. It suffices to show that $C = \mathbb{R}$, so that $\bar{\mathbb{Q}} = C = \mathbb{R}$. Since $\mathbb{Q} \subseteq C$, $(\mathbb{R} \backslash C) \cap \mathbb{Q} = \varnothing$, so $\mathbb{R} \backslash C$ has no rational numbers. But every open set (except $\varnothing$) has a rational number, so $\mathbb{R} \backslash C = \varnothing$, or $C = \mathbb{R}$.

Problem 26.4

Let

$$\mathcal{B} = \{(a, b) : a < b, a, b \in \mathbb{Q}\}.$$

This is a countable basis for $\mathbb{R}$. Let $\{U_\alpha : \alpha \in A\}$ be an open cover of $\mathbb{R}$. Consider the collection

$$B' = \{B \in \mathcal{B} : \exists\, \alpha \in A \text{ where } B \subseteq U_\alpha\}.$$

Given any $x \in \mathbb{R}$, we can pick some α where $x \in U_\alpha$. Since $\mathcal{B}$ is a basis, there is $B \in \mathcal{B}$ where $x \in B \subseteq U_\alpha$, so $B \in B'$. Therefore $\mathcal{B}'$ still covers $\mathbb{R}$, and it's countable (since $\mathcal{B}$ is countable). So given any open cover of $\mathbb{R}$, we can extract a countable subcover.

Problem 26.5

Part (1)

Assume X is normal, which means singletons are closed and for each pair of closed disjoint $C, D \subseteq X$, there's $U \supseteq C$ and $V \supseteq D$ such that $U \cap V = \varnothing$.

First, we show that X is $\mathbf{T_0}$: let $a, b \in X$. Then $\{a\}, \{b\}$ are closed, so there are open $U \supseteq \{a\}$ and $V \supseteq \{b\}$ where $U \cap V = \varnothing$. So $a \in U$ but $b \notin U$, as desired.

Now, we show that X is $\mathbf{T_3}$. Let $x \in X$ and $C \subseteq X$ be closed with $x \notin C$. Since $\{x\}$ is closed, there are open $U \supseteq \{x\}$ and $V \supseteq C$ where $U \cap V = \varnothing$, as desired.

Part (2)

Assume X is regular, so that for all $a, b \in X$, there's U where $a \in U$ and $b \notin U$ or $b \in U$ and $a \notin U$, and for all x and closed $C \not\ni x$, there are open $U \supseteq \{x\}$ and $V \supseteq C$ where $U \cap V = \varnothing$.

To show that X is Hausdorff, let $x_1, x_2 \in X$. There's open U where $x_1 \in U$ but $x_2 \notin U$. Also, there's $U_1 \supseteq U$ and $U_2 \supseteq \{x_2\}$ where $U_1 \cap U_2 = \varnothing$, so $x_1 \in U_1$ and $x_2 \in U_2$ where $U_1 \cap U_2 = \varnothing$, as desired.

Part (3)

We wish to show that if X is Hausdorff then singleton sets are closed. Let $x \in X$. Then for all $y \in X \backslash \{x\}$, there are $U_y^{(1)}, U_y^{(2)}$ where $U_y^{(1)} \cap U_y^{(2)} = \varnothing$. Let $U = \bigcup_{y \neq x} U_y^{(2)}$. Then $x \notin U$ but all $y \neq x$ is in U. Also, all $U_y^{(2)}$ are open, so U is open, and $U = X \backslash \{x\}$. So $\{x\}$ is closed.

Part (4)

Let $a, b \in X$ and assume singleton sets are closed. We have that $X \backslash \{a\}, X \backslash \{b\}$ are open. Let $U = X \backslash \{a\}$; then U is open and $b \in U$ but $a \notin U$.

Problem 26.6

Assume X is $\mathbf{T_4}$. Let $A \subseteq U$ where A is closed and U is open and set $B = X \backslash U$. Then B is closed and $A \cap B = \varnothing$. By $\mathbf{T_4}$, there are disjoint open neighborhoods V, W where $A \subseteq V$, $B \subseteq W$, and $V \cap W = \varnothing$. Since W is an open neighborhood of $B = X \backslash U$, its complement is a closed set containing V. But $X \backslash W \subseteq U$, since $W \supseteq B = X \backslash U$. So, since $X \backslash W$ is closed, $\bar{V} \subseteq X \backslash W \subseteq U$.

Now assume that for all closed $A \subseteq X$ and open $U \supseteq A$, there exists open V with $A \subseteq V \subseteq \bar{V} \subseteq U$. Let A, B be disjoint closed subsets of X. Then $U = X \backslash B$ is an open neighborhood of A. We know there is an open V with $A \subseteq V \subseteq \bar{V} \subseteq U = X \backslash B$. Let $W = X \backslash \bar{V}$. Then W is open, $B \subseteq W$, and $V \cap W = V \cap (X \backslash \bar{V}) = \varnothing$. So X is $\mathbf{T_4}$.

Problem 26.7

Let $(x_1, y_1), (x_2, y_2) \in X \times Y$. Then there are open U_1, U_2, V_1, V_2 where $x_1 \in U_1, x_2 \in U_2, y_1 \in V_1, y_2 \in V_2$ where $U_1 \cap V_1 = U_2 \cap V_2 = \varnothing$. By the product

topology, $U_1 \times V_1, U_2 \times V_2$ are both open in $X \times Y$ and $x_1, y_1 \in U_1 \times V_1$ and $(x_2, y_2) \in U_2 \times V_2$, so $X \times Y$ is Hausdorff.

Problem 26.10

Let X be $\mathbf{T_{2.5}}$ and $x, y \in X$. There are closed neighborhoods $C \ni x, D \ni y$ with $C \cap D = \varnothing$. Let $U = C^\circ$, $V = D^\circ$; we know that U, V are open. Since C, D are closed neighborhoods of x, y, $x \in U$ and $y \in V$. Therefore, X is Hausdorff.

Problem 26.11

First, if X is $\mathbf{T_1}$, singletons are open so X is $\mathbf{T_{0.5}}$.

Now, assume X is $\mathbf{T_{0.5}}$. Let $a, b \in X$. If one of $\{a\}, \{b\}$ are open (call the open one $\{c\}$), then let $U = \{c\}$. We have that $c \in U$ and $(\{a, b\} \backslash \{c\}) \cap U = \varnothing$, so X is $\mathbf{T_0}$. If both $\{a\}, \{b\}$ are closed, let $U = X \backslash \{b\}$. Then $a \in U$ but $b \notin U$, so X is $\mathbf{T_0}$.

Let $X = \{a, b, c\}$ and $\mathcal{T} = \{\varnothing, \{a, b\}, \{a\}, \{b\}, \{a, b, c\}\}$. Then $(X, \mathcal{T})$ is $\mathbf{T_{0.5}}$ but not $\mathbf{T_1}$, since $\{c\}$ is closed but not open.